

Reddit Sentiment and Stock Performance: A Predictive Analysis

Eshan Prakash
eprakash@oxy.edu
Occidental College
Los Angeles, USA

Eduardo Rebollar
rebollar@oxy.edu
Occidental College
Los Angeles, USA

Aaron Good
agood@oxy.edu
Occidental College
Los Angeles, USA

Abstract

The rise of social media platforms over the past decade has fundamentally altered the way investors access, discuss, and share market information. This study examines whether sentiment expressed in Reddit discussions can predict subsequent stock price movements over different time horizons. We collected Reddit posts from three investment-focused subreddits (r/WallStreetBets, r/stocks, r/StockMarket) during Q2 2023 and used VADER sentiment to quantify user sentiment towards 23 publicly traded stocks across five sectors. We then analyzed whether Q2 sentiment predicted Q3 raw, market-adjusted, or sector-adjusted returns. Quarterly cross-sectional analysis revealed no significant relationship between aggregated Q2 sentiment and Q3 returns ($r \approx 0.31$, $p > 0.14$). However, weekly time-series analysis of the five most-discussed stocks demonstrated a significant negative relationship between adjusted returns and weekly sentiment, with high weekly sentiment predicting weaker returns the following week ($r = -0.29$ to -0.33 , $p < 0.05$). These findings suggest that Reddit sentiment functions as a contrarian indicator at short time scales but shows no predictive validity at longer time horizons, challenging assumptions about social media's role in anticipating market movements.

Keywords: sentiment analysis, stock prediction, Reddit, social media finance, behavioral finance

1 Introduction

The democratization of financial information through social media networks has created new channels through which millions of participants and retail investors rapidly communicate, coordinate, and potentially influence market outcomes [1, 2]. Events such as the famous January 2021 GameStop short squeeze, which gained enough widespread attention to reach mainstream media, showed how online coordinated and collective retail investor activity can drive unprecedented volatility in targeted securities [3]. These dynamics raised an important empirical question: *Does online investor sentiment merely reflect and react to existing market conditions or does it genuinely contain predictive information about future stock price movements?*

Traditional finance theory, grounded in the Efficient Market Hypothesis, suggests that asset prices fully reflect all publicly available information and that sentiment-driven

trading represents irrational noise rather than information and should not meaningfully predict returns [4]. However, behavioral finance research has documented numerous ways in which investor psychology systematically influences market outcomes. Sentiment-based theories suggest that when investors become optimistic about a security, increased demand drives prices upward, creating positive momentum [5]. If social media platforms aggregate and amplify investor sentiment, they might serve as leading indicators of price movements.

Previous research on social media networks and financial markets has produced mixed findings. Several studies have documented positive correlations between social media activity and subsequent returns, interpreting this as evidence that online sentiment contains predictive information [6, 7]. Other work has found negative relationships or time-varying effects, suggesting that the relationship may be more complex than a simple positive prediction [8, 9]. The temporal dynamics of sentiment-return relationships remain particularly unclear, with different studies examining different time scales and reaching differing conclusions.

Our present research aims to address this gap by systematically examining Reddit sentiment as a predictive indicator of stock returns over two distinct time horizons:

- (1) A quarterly cross-sectional analysis examining whether aggregated sentiment during Q2 2023 predicts stock performance in Q3 2023 across 23 stocks, testing this relationship using three model specifications with increasing levels of control: raw returns, market-adjusted returns (controlling for overall market trends), and sector-adjusted returns (controlling for both market and sector-specific movements).
- (2) A weekly time series analysis on the five most-discussed stocks to examine short-term predictive relationships and test for bidirectional causality between sentiment and returns.

We originally hypothesized that a higher average sentiment would correspond to increased stock returns at both horizons, indicating that Reddit sentiment functions as a positive predictor of stock price changes rather than merely reflecting concurrent market trends. We first tested this hypothesis using aggregated quarterly data (Study 1), then examined weekly dynamics to understand shorter-term relationships (Study 2). Our results partially contradict this

expectation: sentiment shows no significant quarterly predictive power but functions as a contrarian indicator at the weekly level.

2 Study 1: Quarterly Cross-Sectional Analysis

2.1 Method

2.1.1 Data Collection. We collected our Reddit data from three investment-focused subreddits during Q2 2023 (April 1 – June 30):

- **r/WallStreetBets** (speculative short-term trading),
- **r/stocks** (fundamental analysis and general discussion), and
- **r/StockMarket** (macroeconomic trends and market commentary).

Posts were included if they explicitly mentioned at least one of the 50 target stocks and met subreddit-specific engagement thresholds (median upvotes or median comments), resulting in 23 stocks qualifying for the quarterly analysis.

2.1.2 Stock Selection and Sentiment Analysis. *Reddit Data Collection.* We used the Pullpush.io API to retrieve archival Reddit posts and metadata [10]. We only included posts written in English, posted within the Q2 2023 window, and that explicitly mention stock tickers or company names from our target list.

Stock Selection. We analyzed 50 large-cap stocks across five economic sectors:

Sector	Tickers
Technology	AAPL, MSFT, GOOGL, AMZN, META, NVDA, TSLA, AMD, NFLX, INTC, CRM, ORCL, ADBE, CSCO, UBER
Finance	JPM, BAC, WFC, GS, MS, C, V, MA, AXP, SCHW
Healthcare	JNJ, UNH, PFE, ABBV, LLY, MRK, TMO, CVS, AMGN, BMY
Energy	XOM, CVX, COP, SLB, EOG, OXY, MPC, PSX, VLO, HAL
Aerospace/Defense	LMT, RTX, BA, NOC, GD

These stocks were selected based on market capitalization, liquidity, and the likelihood of being discussed on retail investor platforms.

Sentiment Analysis Tool. Sentiment was quantified using VADER (Valence Aware Dictionary and sEntiment Reasoner), a lexicon-based sentiment analysis tool developed specifically for social media text [11]. VADER produces a compound sentiment score ranging from -1 (extremely negative) to $+1$ (extremely positive), with scores between -0.05 and $+0.05$ considered neutral.

2.1.3 Stock Price Data. Daily stock prices were retrieved from Yahoo Finance using the `yfinance` Python library [12]. Data spanned April 1–September 30, 2023, covering both the

sentiment measurement period (Q2 2023) and the outcome period (Q3 2023). Benchmark returns were calculated for the S&P 500 (SPY) and sector-specific ETFs (XLK for Technology, XLF for Finance, XLV for Healthcare, XLE for Energy, and ITA for Aerospace/Defense).

2.1.4 Data Preprocessing. *Data Filtering.* To ensure data quality, we applied engagement-based filters to retain posts that met or exceeded subreddit-specific engagement thresholds, specifically the median number of upvotes or the median comment count for their respective subreddit. This resulted in the following thresholds:

- r/WallStreetBets: 27 upvotes or 21 comments
- r/stocks: 11 upvotes or 25 comments
- r/StockMarket: 20 upvotes or 15 comments

This filtering ensures that low-quality or low-visibility posts do not distort sentiment signals.

Sentiment Aggregation. For each Reddit post, we combined the post title and body text into a single document and computed the VADER compound sentiment score. For each stock, we calculated the mean sentiment score across all posts mentioning that stock during Q2 2023. This aggregated sentiment score served as an independent variable.

Return Calculation. Stock returns were calculated as the percentage change in adjusted closing price from the end of Q2 (June 30, 2023) to the end of Q3 (September 30, 2023). We computed three performance measures:

$$\text{Raw Return: } \frac{P_{Q3} - P_{Q2}}{P_{Q2}}$$

$$\text{Market-Adjusted Return: Raw Return} - \text{SPY Return}$$

$$\text{Sector-Adjusted Return: Raw Return} - \text{Sector ETF Return}$$

2.1.5 Statistical Analysis. We tested three regression models with increasing levels of control, each using Q2 average sentiment as the independent variable and Q3 returns (raw, SPY-adjusted which controls overall market trends, or sector-adjusted which controls for both market and sector-specific trends) as the dependent variable. All analyses used ordinary least squares regression with Pearson correlation coefficients and two-tailed significance tests at $\alpha = 0.05$.

2.2 Results

2.2.1 Descriptive Statistics. The quarterly analysis included 23 stocks with complete data. The average Q2 2023 sentiment ranged from 0.168 to 0.833 ($M = 0.456$, $SD = 0.182$), which generally indicates positive sentiment with substantial between-stock variation. Q3 2023 raw returns ranged from -11.2% to 9.1% ($M = -2.8\%$, $SD = 5.4\%$), reflecting the broader market downturn during this period (SPY return = -3.3%). Figure 1 shows the distribution of positive, neutral, and negative posts for each stock, with NVDA, TSLA, and AAPL receiving the most discussion volume and exhibiting predominantly positive sentiment.

Reddit Sentiment and Stock Performance:
A Predictive Analysis

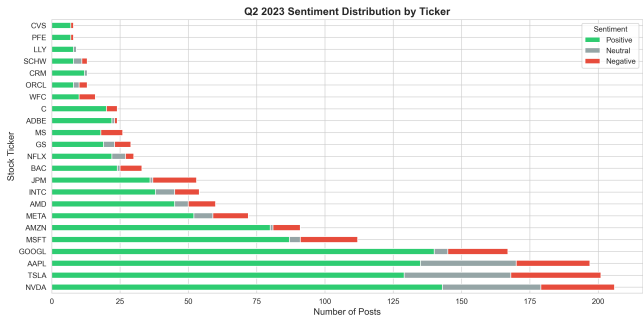


Figure 1. Distribution of sentiment classifications (positive, neutral, negative) across all analyzed stocks during Q2 2023. Bar length represents the total number of posts mentioning each ticker. Stocks are ordered by average sentiment score, with highly positive stocks at the bottom.

2.2.2 Correlation Analysis Summary. We tested three regression models examining the relationship between Q2 sentiment and Q3 returns. Model 1 tested whether sentiment predicted raw returns, and we found the relationship was positive but not statistically significant. Model 2 examined market-adjusted returns (controlling for SPY performance), but because the SPY return was constant across all stocks, this yielded identical results to Model 1. Model 3 examined sector-adjusted returns, controlling for sector-specific trends, and showed a slightly stronger but still non-significant relationship. Table 1 presents the complete results. Across all

Table 1. Quarterly Cross-Sectional Regression Results

Model	Pearson r	p-value	R ²
Model 1: Raw Returns	0.3112	0.1483	0.0968
Model 2: vs SPY	0.3112	0.1483	0.0968
Model 3: vs Sector	0.3153	0.1428	0.0994

three specifications, quarterly sentiment explained less than 10% of variance in subsequent returns and failed to reach conventional levels of statistical significance. The positive direction of effects suggests a weak tendency for stocks with more positive sentiment to outperform, but this relationship was not reliable. Figure 2 visualizes this weak positive relationship for raw returns, with data points color-coded by sector.

Figure 3 compares the correlation coefficients across all three models, demonstrating consistency in the weak positive relationship regardless of which control variables were included.

3 Study 2: Weekly Time Series Analysis

Study 1 revealed no significant predictive relationship between quarterly sentiment and subsequent returns. To examine whether predictive relationships exist at shorter time

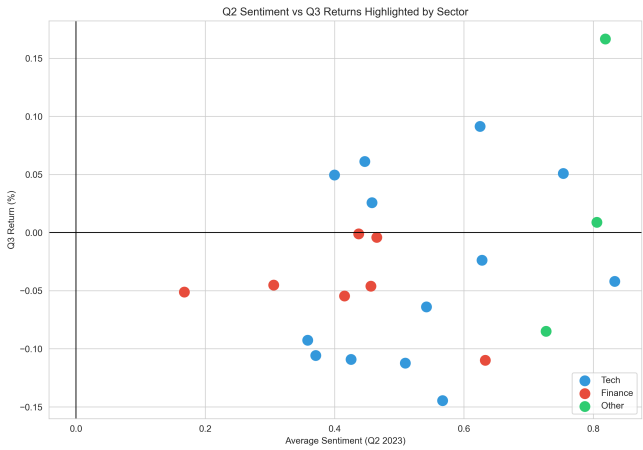


Figure 2. Relationship between Q2 2023 average sentiment and Q3 2023 raw returns across 23 stocks, color-coded by sector. The weak positive correlation ($r = 0.31$, $p = 0.15$) indicates no reliable predictive relationship at the quarterly time scale.

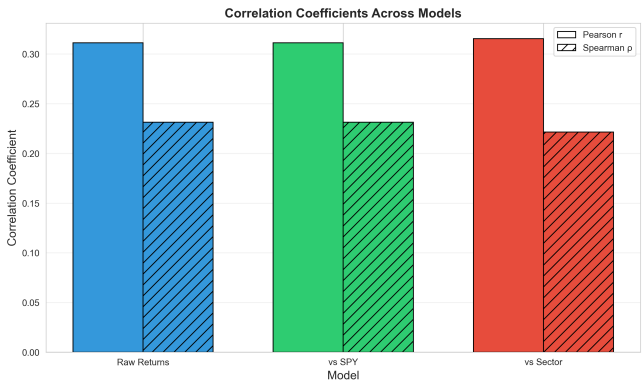


Figure 3. Comparison of Pearson and Spearman correlation coefficients across the three model specifications. All models show consistent weak positive correlations around 0.31, with no significant differences between specifications.

scales, we conducted a weekly time series analysis focusing on the five most-discussed stocks.

3.1 Method

3.1.1 Data Selection. We utilized the same Reddit data collected from Study 1, but focused on the five most-discussed stocks: NVDA, TSLA, AAPL, GOOGL, MSFT. These stocks generated sufficient weekly discussion volume for weekly aggregation, as seen in Figure 4.

3.1.2 Sentiment Analysis and Stock Price Data. The sentiment analysis tool (VADER) and stock price data sources are identical to Study 1.

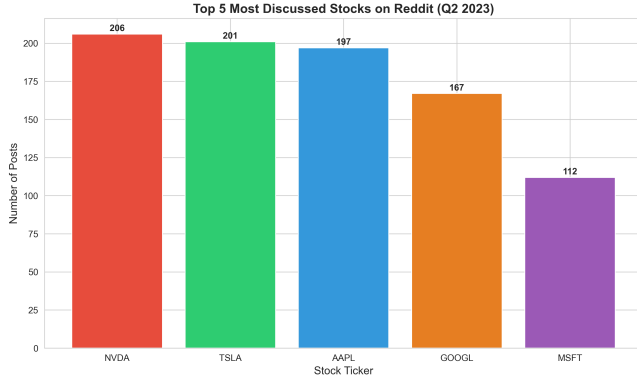


Figure 4. The five most-discussed stocks on Reddit during Q2 2023. NVDA received the most attention with 206 posts, followed by TSLA (201), AAPL (197), GOOGL (167), and MSFT (112). These stocks were selected for weekly time series analysis due to sufficient discussion volume.

3.1.3 Data Preprocessing. Weekly Sentiment. For the five aforementioned stocks, we computed weekly sentiment scores by averaging all VADER compound scores for posts mentioning that stock within each calendar week during Q2 and Q3 2023.

Weekly Returns. Weekly returns were calculated as the percentage change in adjusted closing price from Friday to Friday (or the last trading day of each week), generating three return measures identical to Study 1: raw, market-adjusted (vs SPY), and sector-adjusted returns.

Outlier Removal. To ensure the weekly regression model’s robustness and stability, extreme statistical outliers were removed using a z-score threshold of 3.0. This procedure removed two observations, or approximately two weeks out of sixty (3.3%), from the pooled weekly dataset. The quarterly dataset contained no outliers under the same criterion, so quarterly results remain unchanged.

3.1.4 Statistical Analysis. We conducted weekly lagged regression analysis in two directions using pooled data across the five tickers (approximately 60 weekly observations before outlier removal). Direction 1 tested whether sentiment in week t predicted returns in week $t+1$ (sentiment leading returns). Direction 2 tested whether returns in week t predicted sentiment in week $t+1$ (returns leading sentiment).

1. **Sentiment(t) $\rightarrow$ Returns($t+1$)**
2. **Returns(t) $\rightarrow$ Sentiment($t+1$)**

Analyses used ordinary least squares regression with Pearson correlation coefficients and two-tailed significance tests at $\alpha = 0.05$.

3.2 Results

3.2.1 Direction 1: Sentiment(t) Predicts Returns($t+1$). In contrast to the quarterly findings from Study 1, weekly

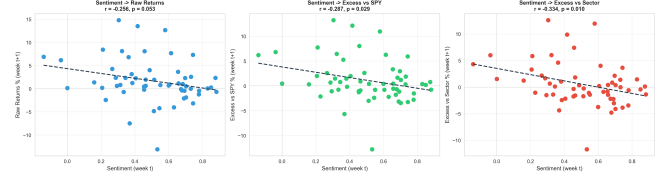


Figure 5. Relationship between weekly sentiment and next week’s returns across three specifications (raw returns, market-adjusted, sector-adjusted) for the five most-discussed stocks. Negative correlations indicate that high sentiment predicts weaker subsequent performance, particularly for adjusted returns.

sentiment demonstrated significant predictive power for next week’s returns when controlling for market and sector movements. After removing outliers, stocks with higher sentiment in week t showed weaker returns in week $t + 1$ across the adjusted return measures. As shown in Table 2, for raw returns, the relationship weakened and became marginally non-significant, for market-adjusted returns, the relationship was significant, and for sector-adjusted returns, the relationship was significantly the strongest.

Table 2. Weekly Sentiment Predicting Next Week’s Returns

Return Type	Pearson r	p-value	Result
Raw Returns	-0.256	0.053	Not Sig.
vs SPY	-0.287	0.029	Significant
vs Sector	-0.334	0.010	Significant

The negative coefficients indicate that weeks characterized by high Reddit sentiment were followed by weeks of relatively poor performance, especially once market and sector trends were accounted for. Figure 5 displays these relationships visually across all three return measures, with clear negative slopes evident in each panel.

Figure 6 presents the temporal dynamics of sentiment and price movements for each of the five stocks, illustrating how sentiment and returns often move in opposite directions or show temporal lags.

3.2.2 Direction 2: Returns(t) Predict Sentiment($t+1$). We also tested the reverse direction, examining whether returns in week t predicted sentiment in week $t+1$. As shown in table 3 after outlier removal, this relationship remained positive but did not reach statistical significance across any return measure.

While not statistically significant (all $p > 0.05$), the positive direction suggests that price increases may lead to somewhat more positive subsequent discussion, though this effect was weak and unreliable. Figure 7 displays these relationships visually across all three return measures, showing weak positive slopes that do not reach statistical significance.

Reddit Sentiment and Stock Performance:
A Predictive Analysis

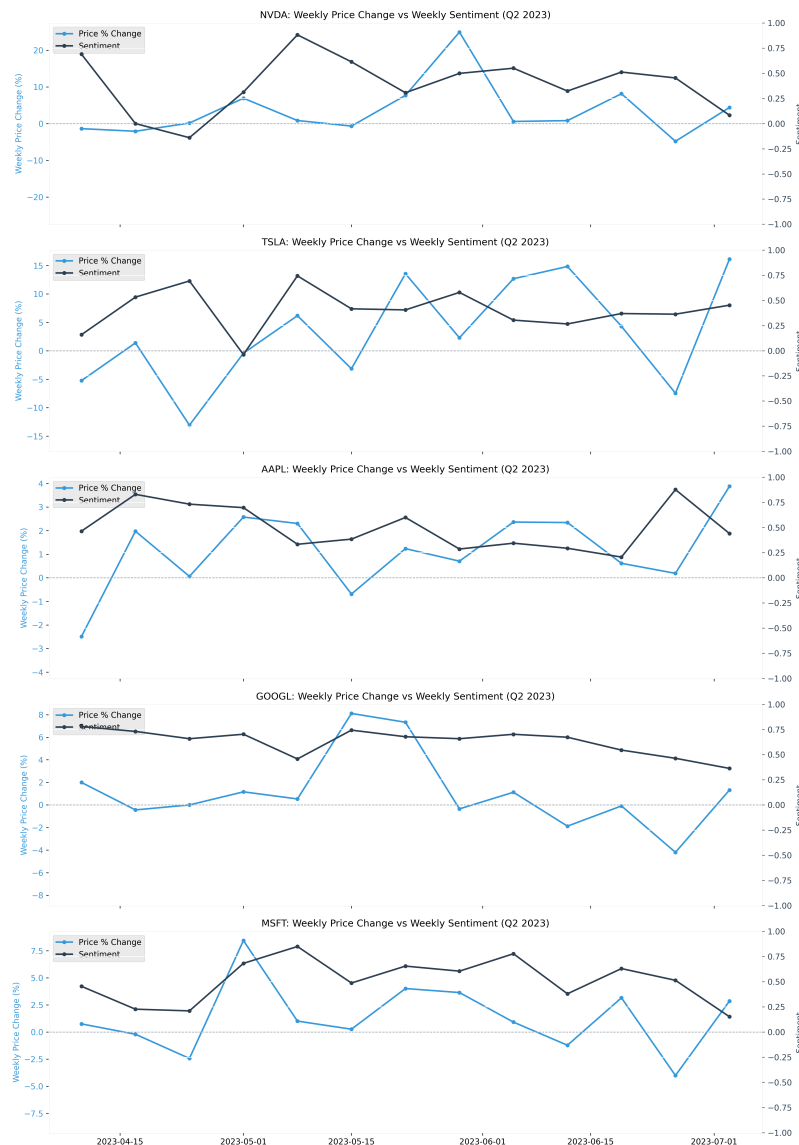


Figure 6. Weekly price changes (blue line, left axis) and sentiment scores (black line, right axis) for the five most-discussed stocks during Q2 2023. The temporal misalignment between sentiment peaks and subsequent price movements illustrates the contrarian nature of the relationship.

Table 3. Weekly Returns Predicting Next Week’s Sentiment

Return Type	Pearson r	p-value	Result
Raw Returns	+0.258	0.051	Not Sig.
vs SPY	+0.254	0.055	Not Sig.
vs Sector	+0.172	0.196	Not Sig.

Figure 8 compares the statistical significance of both directional relationships, illustrating that sentiment reliably predicts returns for adjusted measures while returns do not reliably predict sentiment.

4 Discussion

The present findings reveal a complex, time-dependent relationship between Reddit sentiment and stock returns that challenges simple interpretations of social media as either a leading or lagging indicator of market movements. Our results demonstrate that the predictive validity of sentiment depends critically on the temporal horizon examined, with opposite patterns emerging at weekly versus quarterly time scales.

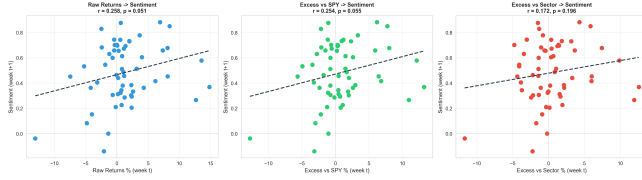


Figure 7. Relationship between weekly returns and next week's sentiment across three specifications (raw returns, market-adjusted, sector-adjusted). All three panels show weak positive but non-significant correlations, indicating that price movements do not reliably predict subsequent sentiment.

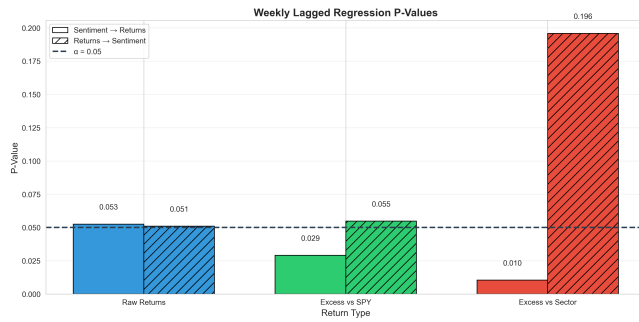


Figure 8. Comparison of p-values for bidirectional lagged regressions after outlier removal. Solid bars show p-values for sentiment predicting next week's returns (two of three models below $\alpha = 0.05$). Hatched bars show p-values for returns predicting next week's sentiment (all above $\alpha = 0.05$). The horizontal dashed line marks the conventional significance threshold.

4.1 Interpretation of Results

At the quarterly level, we observed no significant relationship between aggregated Q2 sentiment and Q3 returns, despite testing multiple specifications that controlled for market and sector effects. While the positive direction of effects ($r \approx 0.31$) is consistent with theories suggesting sentiment captures investor enthusiasm that drives future demand, the magnitude was small and statistically unreliable. This null result contrasts with some prior research documenting positive sentiment-return relationships, aligning with efficient market perspectives and suggesting that publicly available sentiment information becomes rapidly incorporated into prices. The quarterly aggregation period may be too long for sentiment to retain predictive power, as information that drove initial sentiment likely becomes reflected in prices within shorter windows.

In contrast, the weekly time series analysis revealed a striking pattern: high sentiment in week t predicted significantly weaker returns in week $t+1$, particularly after adjusting for market and sector trends. The relationship for raw returns

weakened after outlier removal and became marginally non-significant ($p = 0.053$), though the directional consistency with the adjusted models suggests that the underlying effect is robust but sensitive to extreme observations. These patterns reflect speculative excitement that temporarily inflates prices beyond fundamental values, leading to subsequent corrections or mean reversion. They indicate that weekly sentiment operates as a contrarian indicator rather than a leading indicator of positive performance and also suggest the relationship primarily flows from sentiment to returns rather than vice versa.

The contrarian pattern at weekly horizons directly contradicts the positive (though non-significant) relationship observed at quarterly horizons. This reversal suggests that short-term sentiment spikes represent noise or investor over-reaction, while longer-term sustained sentiment may weakly capture genuine investor interest. However, neither horizon produced strong predictive relationships, suggesting Reddit sentiment has limited practical utility for forecasting returns.

4.2 Practical Implications

These findings have several implications for investors and market participants. First, investors should be cautious about interpreting surges in positive Reddit sentiment as bullish signals for near-term performance. Our results suggest the opposite: stocks experiencing unusual positive attention on Reddit tend to underperform in subsequent weeks, possibly due to temporary overvaluation or profit-taking following sentiment-driven rallies. Second, sentiment aggregated over longer periods shows no reliable predictive relationship with subsequent returns, suggesting quarterly sentiment trends should not be used as standalone investment signals. Third, the contrarian pattern at weekly scales suggests that sentiment extremes may serve as risk indicators rather than opportunity signals, potentially useful for identifying overheated securities.

4.3 Limitations and Future Directions

Several limitations qualify these findings. First, the sample size was relatively small (23 stocks for quarterly analysis, 5 for weekly analysis) and focused exclusively on large-cap stocks frequently discussed on Reddit. The sentiment-return relationship may differ for smaller, less liquid securities, where retail investor coordination could have a stronger price impact. Second, the study examined a single three-month period during relatively stable market conditions. The predictive validity of sentiment may vary across different market regimes, particularly during periods of extreme volatility or crisis. Third, VADER sentiment analysis, while validated for social media text, may not capture the nuanced financial language, sarcasm, or irony common in communities like r/WallStreetBets. More sophisticated natural language processing approaches might yield different

results. In support of this concern, a robustness check comparing VADER with a finance-specific RoBERTa [13] model showed only weak agreement between the two systems (approximately $r = 0.24$ correlation and 42% label agreement), indicating that sentiment classification on Reddit is highly model-dependent. This suggests that some of our findings may reflect VADER's lexicon-based assumptions rather than a stable underlying sentiment signal.

Fourth, our analysis did not control for potential confounding variables such as earnings announcements, analyst revisions, news events, or institutional trading activity that might drive both sentiment and returns. Fifth, we examined only linear relationships at one-week and one-quarter lags. Nonlinear effects or different lag structures might reveal alternative patterns. Finally, the observational design cannot definitively establish causal relationships. While the temporal precedence of sentiment before returns is consistent with sentiment influencing prices, alternative explanations (such as both variables responding to unobserved third factors) remain possible.

Future research should address these limitations by examining larger samples of stocks, including small and mid-cap securities, analyzing multiple time periods and market conditions, employing more sophisticated sentiment analysis techniques, controlling for fundamental and technical factors, and exploring nonlinear and time-varying relationships. Additionally, experimental or quasi-experimental designs could help establish causal mechanisms underlying sentiment-return relationships.

4.4 Conclusion

This study investigated whether Reddit sentiment predicts future stock returns across quarterly and weekly time horizons. Quarterly aggregated sentiment showed no significant predictive relationship with subsequent returns ($r \approx 0.31$, $p > 0.14$), suggesting that sustained discussion about a stock over three months does not reliably forecast its next-quarter performance. In contrast, weekly sentiment demonstrated significant negative predictive power for adjusted returns, with high sentiment weeks followed by weaker subsequent returns ($r \approx -0.29$ to -0.33 , $p < 0.05$). The relationship for raw returns became marginally non-significant after outlier removal, though the directional consistency persists. This contrarian pattern suggests that short-term sentiment spikes on Reddit may reflect overoptimism or speculative excess that precedes price corrections rather than signaling genuine opportunities.

The time-dependent reversal in the sentiment-return relationship—positive but non-significant at quarterly scales, negative and significant for weekly adjusted returns—demonstrates that the predictive validity of social media sentiment strongly depends on temporal aggregation. These findings challenge simple interpretations of social media as either a consistently leading or lagging indicator and highlight

the importance of considering multiple time horizons when studying sentiment-driven market dynamics. While Reddit sentiment shows limited practical utility for forecasting returns, its contrarian properties at short time scales suggest it may serve as a useful indicator of potential overvaluation or speculative excess in heavily discussed securities.

Acknowledgments

We acknowledge and thank the developers of the Pullpush.io API for enabling access to archival and historical Reddit data, the creators and maintainers of the VADER sentiment analysis toolkit, the creators of the financial-RoBERTa sentiment model used in our robustness checks, and the developers of the open-source Python scientific computing ecosystem, including: pandas, NumPy, matplotlib, seaborn, SciPy, scikit-learn, yfinance, transformers, torch, and safetensors. All data collection and analysis were conducted in accordance with Reddit's API terms of service and research guidelines. All code and data processing scripts are available at <https://github.com/EshanPrakash/reddit-stock-sentiment>.

References

- [1] S. Warkulat, and M. Pelster. Social media attention and retail investor behavior: Evidence from r/wallstreetbets, *International Review of Financial Analysis*, Volume 96, Part B, 2024, 103721, <https://doi.org/10.1016/j.irfa.2024.103721>
- [2] M. Farrell, T.C. Green, R. Jame, and S. Markov. The democratization of investment research and the informativeness of retail investor trading, *Journal of Financial Economics*, Volume 145, Issue 2, Part B, 2022, Pages 616-641, <https://doi.org/10.1016/j.jfineco.2021.07.018>
- [3] S. Long, B. Lucey, Y. Xie, and L. Yarovaya. "I just like the stock": The role of Reddit sentiment in the GameStop share rally, *Finance Research Letters*, Volume 58, Issue 1, 2022, Pages 19-37, <https://doi.org/10.1111/fire.12328>
- [4] B.G. Malkiel, and E.F. Fama. EFFICIENT CAPITAL MARKETS: A REVIEW OF THEORY AND EMPIRICAL WORK*, *The Journal of Finance*, Volume 25, Issue 2, 1970, Pages 383-417, <https://www.jstor.org/stable/2325486>
- [5] B.P. Malcoln, and W. Jeffrey. Investor Sentiment and the Cross-Section of Stock Returns, *The Journal of Finance*, Volume 61, Issue 4, 2004, Pages 1645-1680, <https://doi.org/10.1111/j.1540-6261.2006.00885.x>
- [6] B. Johan, M. Huina, and Z. Xiaojun. Twitter mood predicts the stock market, *Journal of Computational Science*, Volume 2, Issue 1, 2011, Pages 1-8, <https://doi.org/10.1016/j.jocs.2010.12.007>
- [7] H.K. Sul, A.R. Dennis, L. Yuan. Trading on Twitter: The Financial Information Content of Emotion in Social Media, *2014 47th Hawaii International Conference on System Sciences*, 2014, Pages 806-815, <https://api.semanticscholar.org/CorpusID:1893868>
- [8] T. Renault. Market manipulation and suspicious stock recommendations on social media, *SSRN Electric Journal*, 2017, 10.2139, <http://dx.doi.org/10.2139/ssrn.3010850>
- [9] D. Garcia. Sentiment During Recessions, *The Journal of Finance*, Volume 68, Issue 3, 2013, Pages 1267-1300 <https://dx.doi.org/10.2139/ssrn.1571101>
- [10] Pullpush.io. Reddit Archival Data API. <https://pullpush.io/>
- [11] C.J. Hutto, and E. Gilbert. VADER: A Parsimonious Rule-based Model for Sentiment Analysis of Social Media Text. *Proceedings of the Eighth International AAAI Conference on Weblogs and Social Media*, 2014, 216-225.

- [12] Ran Aroussi. yfinance: Download market data from Yahoo! Finance API. <https://pypi.org/project/yfinance/>
- [13] Mohammad Soleimanian. financial-roberta-large-sentiment: A RoBERTa model fine-tuned on financial text for sentiment analysis. <https://huggingface.co/soleimanian/financial-roberta-large-sentiment>